

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Randomized, Open-Label Study Evaluating the Efficacy and Safety of Inavolisib Plus Fulvestrant Versus Alpelisib Plus Fulvestrant in Patients With Hormone Receptor-Positive, HER2-Negative, PIK3CA Mutated, Locally Advanced or Metastatic Breast Cancer Who Progressed During or After CDK4/6 Inhibitor and Endocrine Combination Therapy **Short Title/Acronym:** INAV0121 **Protocol Number:** W043919 **NCT Number:** NCT05646862 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Inavolisib plus fulvestrant (Active Comparator: Alpelisib plus fulvestrant) **Phase of Development:** Phase III **Trial Status:** Active, Not Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of the targeted PI3K α inhibitor inavolisib combined with fulvestrant compared with alpelisib plus fulvestrant by measuring Progression-Free Survival (PFS) in patients with HR-positive, HER2-negative, PIK3CA-mutated, locally advanced (LA) or metastatic breast cancer (mBC). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Best Overall Response (BOR), Duration of Response (DoR), Clinical Benefit Rate (CBR), safety, tolerability, patient-reported outcomes, and pharmacokinetics (PK). **Sub-study Objective:** To evaluate the impact of repeat doses of inavolisib (coadministered with fulvestrant) on single-dose pharmacokinetics of sensitive CYP450 enzyme substrates (midazolam, omeprazole, and bupropion) to assess drug-drug interactions (DDI).

2.2 Background & Rationale Dysregulating mutations in the *PIK3CA* gene occur in approximately 40% of HR-positive, HER2-negative breast cancers, representing a common mechanism of resistance to standard cyclin-dependent kinase 4/6 inhibitors (CDK4/6i) and endocrine therapy (ET) combinations. While alpelisib (a PI3K α inhibitor) plus fulvestrant is an approved treatment in this setting, its widespread clinical implementation can be challenging due to tolerability issues. Inavolisib is a highly potent, selective PI3K α inhibitor that also facilitates the degradation of the mutated PI3K α isoform. This study is designed to evaluate if inavolisib plus fulvestrant can provide a better therapeutic index and improved outcomes compared to the standard of care alpelisib regimen.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Main Efficacy Study + DDI PK Sub-study) **Control Type:** Active Comparator (Alpelisib + Fulvestrant) **Blinding:** Open-label (with Blinded Independent Central Review [BICR] for endpoints) **Randomization:** Randomized, 1:1 ratio

3.2 Planned Enrollment & Duration Target **\$\$\$:** 420 evaluable patients **Number of Sites:** ~200 sites globally **Estimated Study Start:** 07-Jun-2023 **Estimated Primary Completion:** 06-Apr-2026 (Estimated Study Completion: 30-Mar-2029)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed adenocarcinoma of the breast that is locally advanced (LA) or metastatic (mBC) and is not amenable to surgical or radiation therapy with curative intent. 2. Documented HR-positive / HER2-negative tumor according to ASCO/CAP guidelines with confirmation of *PIK3CA* mutation(s) via a specified test (circulating-tumor DNA or tumor tissue). 3. Disease progression after or during treatment with a combination of a CDK4/6 inhibitor and endocrine therapy (≤ 2 prior lines of systemic therapy in the mBC setting; one line of chemotherapy allowed).

4.2 Exclusion Criteria 1. Metaplastic breast cancer. 2. Type 2 diabetes requiring ongoing systemic treatment at the time of study entry, or any history of Type 1 diabetes. 3. Inability or unwillingness to swallow pills, or malabsorption syndrome interfering with enteral absorption. 4. Any history of leptomeningeal disease, carcinomatous meningitis, or known active/untreated central nervous system (CNS) metastases. 5. Known active, systemic infection at study enrollment, or any major episode of infection requiring treatment with intravenous antibiotics within 7 days prior to Day 1 of Cycle 1. 6. Active inflammatory or infectious conditions in either eye, or a history of idiopathic/autoimmune-associated uveitis. 7. *Sub-study Specific Exclusion:* Treatment with mild, moderate, or strong inducers or inhibitors of CYP2B6, CYP3A4, and/or CYP2C19 (e.g., St. John's Wort, grapefruit juice) within 14 days or 5 drug-elimination half-lives prior to initiation of study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Inavolisib:* 9 mg tablet orally once a day (PO QD) on Days 1-28 of each 28-day cycle. * *Fulvestrant:* 500 mg administered on Days 1 and 15 of Cycle 1, and then on Day 1 of each subsequent 28-day cycle. (Comparator arm receives Alpelisib + Fulvestrant based on standard prescribing intervals). **Sub-study Agents:** Single doses

of sensitive CYP450 enzyme substrates (midazolam, omeprazole, and bupropion) at predefined intervals for PK evaluation. **Route of Administration:** Oral (inavolisib/alpelisib/sub-study agents) and Intramuscular/Systemic (fulvestrant). **Duration of Treatment:** Administered in 28-day cycles until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Luteinizing hormone-releasing hormone (LHRH) agonist therapy for pre/perimenopausal women and men (must begin at least 2 weeks prior to Day 1 of Cycle 1). **Prohibited:** Other concurrent systemic anti-cancer therapies outside of the assigned protocol parameters.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, Biomarker/Mutation Testing (PIK3CA, HR/HER2 status), Baseline Scans
Baseline	Day 1 (Cycle 1)	Randomization, First Dose of Inavolisib + Fulvestrant (or Alpelisib + Fulvestrant)
Interim	Q28 Days	Ongoing cycles: Safety Labs, PK sampling (incl. sub-study DDI sampling for midazolam/bupropion/omeprazole), Efficacy/Tumor Assessments (RECIST v1.1)
End of Study	Variable	Disease progression, unacceptable toxicity exit, Safety Follow-up, Survival tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Main Study:** Progression-Free Survival (PFS), defined as the time from randomization to the time when the disease progresses or a patient dies from any cause. Evaluated via Blinded Independent Central Review (BICR) using RECIST v1.1 guidelines. * **Sub-study:** Maximum observed Drug Concentration (Cmax) for Midazolam, and Cmax for Bupropion.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Objective Response Rate (ORR) by BICR * Best Overall Response (BOR) by BICR * Duration of Response (DoR) and Clinical Benefit Rate (CBR) by BICR * Safety, tolerability, patient-reported outcomes, and pharmacokinetics (PK)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous safety and tolerability monitoring standard for PI3Kα inhibitors (including assessment for hyperglycemia, rash, and gastrointestinal events). **Serious Adverse Events (SAEs):** Standard oncology trial safety reporting (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) is in place for evaluating ongoing safety and efficacy.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (PFS/OS). Safety Set for AE/SAE evaluations.

Sub-study PK population for DDI evaluation. **Sample Size**

Justification: \$N=420\$ patients powered to detect a statistically significant difference in PFS. **Handling of Missing Data:** A stratified log-rank test at an overall 0.05 significance level (two-sided) will be used for the primary endpoint. Median PFS will be estimated using Kaplan-Meier methodology with appropriate right-censoring for missing/lost-to-follow-up data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Continuous site monitoring across the ~200 global locations via Sponsor Clinical Operations. **Audit Trail:** eCRF locking, query resolution via data clarification forms, and blinded independent central radiological reviews for unbiased endpoint validation.

11. Study Identifiers & Governance

Primary ID: W043919 **Registry Number:** NCT05646862 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** INAV0121 **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Open-Label Study Evaluating the Efficacy and Safety of Inavolisib Plus Fulvestrant Versus Alpelisib Plus Fulvestrant in Patients With Hormone Receptor-Positive, HER2-Negative, PIK3CA Mutated, Locally Advanced or Metastatic Breast Cancer Who Progressed During or After CDK4/6 Inhibitor and Endocrine Combination Therapy

12. Clinical Framework (Breast Cancer Specific)

Disease Area: Breast Cancer **Primary Condition:** Locally Advanced or Metastatic Breast Cancer (LA/mBC) **Diagnosis Threshold:** HR-Positive, HER2-Negative, with documented *PIK3CA* Mutation **Medical Methodology:** PI3K α Targeted Therapy + Endocrine Therapy **Combination Optimization Target:** Delayed disease progression / Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Targeted Oncology / Post-CDK4/6i Resistance Pathway **Education Protocol (HCP):** Global investigator meetings,

protocol training for AE management (e.g., stomatitis, hyperglycemia) **Education Protocol (Patient):** Informed consent, oral medication adherence counseling, toxicity monitoring **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for tumor scans & Independent Data Monitoring Committee (IDMC)

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until study completion (Est. 2029) or death **Onsite Visit Cadence:** Day 1 of every 28-day cycle (+ Day 15 in Cycle 1) **Remote Monitoring Frequency:** Standard ongoing oncology surveillance between clinical dosing **Checklist Review Interval:** Regular tumor imaging according to RECIST v1.1 guidelines

15. Sign-off & Approval

Role	Name	Signature	Date		:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					